

SurgRFO: Foundation Model Based Compositional Synthesis of Critical Retained Foreign Objects in Intraoperative Chest X-rays

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1. Background / Motivation

- Critical retained foreign objects (RFOs)** such as surgical sponges and needle fragments are rare but high-risk patient safety events.
- Intraoperative chest radiographs are visually complex cluttered scenes, overlapping devices, variable anatomy exposure, and heterogeneous acquisition settings.
- Positive RFO cases are extremely scarce**, limiting robust detection model training and generalization.
- Goal:** generate realistic RFO-positive intraoperative chest X-rays to augment training for critical RFO detection.

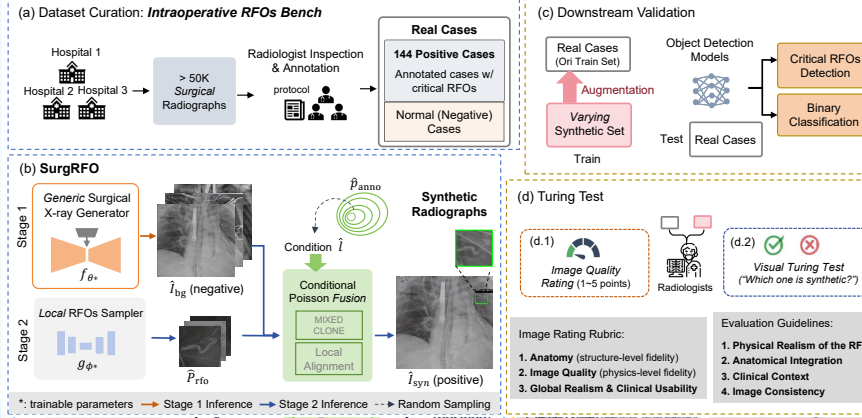
2. Key Contributions

- Introduced **SurgRFO**, a two-stage synthesis framework that separates global background generation from local RFO generation.
- Released **Intraoperative RFO Bench**, the first publicly available de-identified benchmark dedicated to critical RFOs in intraoperative chest radiographs.
- Demonstrated improved downstream detection across Faster R-CNN, RetinaNet, and YOLOv8.
- Validated realism with a radiologist-driven Turing test.

3. Dataset / Study Design

>50K surgical radiographs screened	JHH: 144 critical RFO-positive cases
JHH: 944 negative cases	CU external cohort: 20 critical RFO-positive cases
18 years of surgical radiographs from two health systems	Patient-level split: 6:1:3 for training / validation / test

4. Method: SurgRFO Overview



Overview of the Intraoperative RFO Bench and SurgRFO pipeline. (a) Multi-institutional intraoperative radiographs are curated and expert-annotated. (b) SurgRFO applies a two-stage synthesis framework, combining surgical X-ray background generation with localized RFO sampling and fusion to create synthetic images. (c) Synthetic data are used for training augmentation and evaluated on held-out real cases. (d) A radiologist-led Turing test assesses visual realism.

5. Representative Synthesis Results

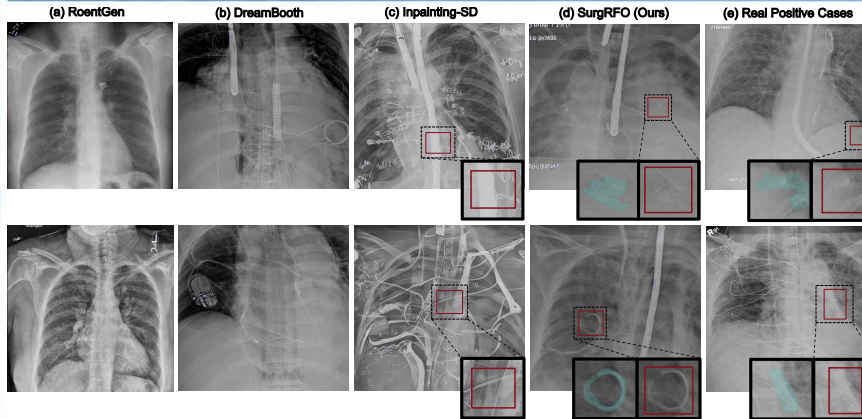


Fig. 2. Visualization of synthesis results. Columns (a-b) show Stage-1 RFO-negative background generation using the original RoentGen and DreamBooth fine-tuning. Columns (c-d) compare Stage-2 RFO-positive synthesis using inpainting Stable Diffusion versus the proposed SurgRFO framework, with real positive cases shown in (e) for reference.

6. Quantitative Synthesis Quality

Stage 1 RFO-negative synthesis

Method	FID ↓	MS-SSIM ↑
Original ReontGen SD	175.167	0.262 ± 0.130
DreamBooth SD	80.963	0.424 ± 0.085
Ours (best at 2k)	43.675	-
Ours (best diversity at 4k)	-	0.362 ± 0.054

SurgRFO achieves the best background fidelity and strong diversity.

Stage 2 RFO-positive synthesis

Method	FID ↓	MS-SSIM ↑
Original ReontGen SD	197.837	0.262 ± 0.130
Inpainting SD	154.523	0.118 ± 0.045
Ours	69.178	0.480 ± 0.064

SurgRFO substantially improves fidelity over positive-image baselines.

7. Downstream Detection Gains

Detection on held-out real radiographs

Method	mAP@0.3 ↑	FNR ↓	FROC ↑	ACC ↑
Faster R-CNN (base → best)	0.184 → 0.510	78.79% → 33.33%	0.242 → 0.636	0.672 → 0.954
RetinaNet (base → best)	0.099 → 0.564	72.73% → 36.36%	0.455 → 0.727	0.921 → 0.954
YOLOv8 (base → best)	0.000 → 0.357	100.00% → 60.61%	 → 0.394	0.892 → 0.931

- Synthetic augmentation consistently improves sensitivity and overall retrieval performance.
- Benefits transfer to an external cohort, suggesting improved robustness under dataset shift.

8. Radiologist-driven Turing Test

Blinded image quality ratings showed no significant difference between real and synthetic images.

- Rater 1: real 2.68 ± 1.11 vs synthetic 2.47 ± 1.02 , $p = 0.166$
- Rater 2: real 2.64 ± 1.22 vs synthetic 2.31 ± 0.74 , $p = 0.220$
- Discrimination accuracy: 0.750 for both raters
- Synthetic → Real rate: 0.30 (Rater 1), 0.25 (Rater 2)
- Synthetic images were often judged visually plausible and sometimes mistaken as real.

9. Conclusion / Take-home Message

- ✓ SurgRFO addresses critical RFO data scarcity with structured two-stage synthesis.
- ✓ Synthetic data augmentation improves critical RFO detection across multiple detector families.
- ✓ Radiologist evaluation supports the clinical plausibility of generated images.

The code is publicly available on GitHub at <https://github.com/Yuanyun936/SurgRFO>